

Business Understanding Report

1. Introduction

1.1 Purpose of the Document

This document presents the Business Understanding phase of the Retail Business Intelligence Platform project based on the Olist Brazilian E-Commerce Dataset. The purpose of this phase is to understand the business environment, identify organizational challenges, define business objectives, and establish the foundation

1.2 Project Overview

The project aims to design and develop a complete Business Intelligence platform capable of transforming raw operational data into valuable business insights. The solution will include data integration, data warehousing, ETL processes, analytical dashboards, and predictive analytics to support strategic and operational decision-making.

2. Company Overview

2.1 Company Background

Olist is a Brazilian e-commerce marketplace that connects thousands of sellers with customers through major online marketplaces. The company enables small and medium-sized businesses to sell products online while managing logistics, payments, customer orders, and product information.

The company generates a significant amount of transactional data every day, including customer information, orders, products, payments, deliveries, and customer reviews.

2.2 Business Model

Olist operates as an intermediary between sellers and customers by providing an integrated platform that manages the complete sales process.

Its core business activities include:

- Customer Management
 - Seller Management
 - Product Management
 - Order Processing
 - Payment Management
 - Logistics Coordination
 - Customer Review Management
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3. Business Context

As Olist continues to grow, the amount of operational data generated increases rapidly. Although the company stores large volumes of data, decision-makers lack a centralized analytical platform capable of transforming this information into meaningful insights.

Business reports are generated from multiple operational datasets, making analysis time-consuming and reducing decision-making efficiency.

To improve business performance, Olist requires a modern Business Intelligence platform that centralizes data, automates reporting, and provides real-time insights.

4. Problem Statement

The current business environment presents several challenges:

- Data is distributed across multiple CSV files.
- There is no centralized reporting platform.

- Manual reporting consumes significant time.
- Business performance is difficult to monitor.
- Customer behavior is not fully analyzed.
- Seller performance lacks standardized evaluation.
- Delivery performance is not continuously monitored.
- Strategic decisions rely on fragmented information.

As a result, managers face difficulties in identifying trends, measuring business performance, and making informed decisions.

5. Project Objectives

5.1 General Objective

Design and implement a complete Business Intelligence platform that transforms operational data into strategic information for business decision-making.

5.2 Specific Objectives

The project aims to:

- Collect and integrate operational data.
 - Build a centralized Data Warehouse.
 - Design ETL processes using Talend.
 - Improve data quality through cleansing and validation.
 - Develop interactive Power BI dashboards.
 - Monitor business KPIs.
 - Perform customer and sales analytics.
 - Build predictive Machine Learning models.
 - Automate data refresh processes using n8n.
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6. Project Scope

In Scope

The project includes:

- Data collection from Olist datasets.
- Data profiling.
- Data cleaning.
- PostgreSQL database implementation.
- Staging Area design.
- Data Warehouse implementation.
- Star Schema modeling.
- ETL development using Talend.
- SQL analytics.
- Power BI dashboard development.
- Machine Learning analysis.
- Workflow automation using n8n.
- Technical documentation.

Stakeholder	Responsibilities	Information Required
Chief Executive Officer (CEO)	Strategic decision-making	Overall company performance
Sales Manager	Sales monitoring	Revenue, sales growth, top products
Marketing Manager	Customer analysis	Customer segmentation, purchasing behavior
Operations Manager	Logistics monitoring	Delivery performance, shipping delays
Finance Manager	Financial monitoring	Revenue, payments, financial KPIs
Business Intelligence Team	BI platform development	Data integration, reporting, analytics

8. Business Processes

The primary business workflow is illustrated below:

Customer



Browse Products



Place Order



Payment Processing



Seller Preparation



Shipping



Delivery



Customer Review



Business Analytics

9. Business Questions

The Business Intelligence platform should answer key business questions such as:

Sales Analysis

- What is the total revenue?
- Which products generate the highest sales?
- Which product categories perform best?

- What are the monthly sales trends?

Customer Analysis

- Who are the most valuable customers?
- What is the customer purchase frequency?
- Which regions generate the highest number of customers?

Product Analysis

- Which products sell the most?
- Which products have poor performance?
- Which categories contribute the highest revenue?

Seller Analysis

- Who are the top-performing sellers?
- Which sellers generate the highest revenue?

Logistics Analysis

- What is the average delivery time?
- How many deliveries are delayed?
- Which regions experience longer delivery times?

Customer Satisfaction

- What is the average review score?
- Does delivery time affect customer satisfaction?

10. Success Criteria

The project will be considered successful if:

- A centralized Business Intelligence platform is implemented.
- All required KPIs are available.
- Dashboards provide interactive and meaningful insights.
- ETL processes operate successfully.
- Decision-makers can monitor business performance efficiently.
- Predictive analytics improve business forecasting.

11. Risks and Constraints

Potential project risks include:

- Missing values in the dataset.
 - Duplicate records.
 - Data quality issues.
 - Limited historical data (2016–2018).
 - Absence of inventory information.
 - Absence of product cost data.
 - Performance limitations when processing large datasets.
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12. Expected Benefits

The proposed Business Intelligence platform will provide several business benefits:

- Faster decision-making.
 - Centralized business reporting.
 - Improved sales monitoring.
 - Better customer understanding.
 - Enhanced seller performance evaluation.
 - Improved operational efficiency.
 - Automated reporting processes.
 - Data-driven strategic planning.
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13. Deliverables

The project deliverables include:

- Business Understanding Report
- Project Charter
- Business Requirements Document
- Source System Analysis
- Data Profiling Report
- PostgreSQL Database
- Staging Area

- Data Warehouse
 - Star Schema
 - ETL Workflows (Talend)
 - SQL Scripts
 - Power BI Dashboards
 - Machine Learning Models
 - n8n Automation Workflows
 - Technical Documentation
 - GitHub Repository
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14. Conclusion

The Business Understanding phase establishes the business foundation of the Retail Business Intelligence Platform project. It identifies the organization's objectives, business challenges, stakeholders, and analytical requirements that will guide the technical implementation of the solution.

The next phase, **Project Charter**, will formally define the project scope, timeline, milestones, resources, assumptions, constraints, and governance, ensuring that the project is managed according to professional Business Intelligence project management practices.